

«DEVELOPMENT OF A SYNTACTIC ANALYZER»

Problem Statement

1. Develop a syntax analyzer (SA) program for a subset of the SIGNAL programming language according to the variant's grammar.
2. The program must provide the following:
 - reading the token stream and tables generated by the lexical analyzer developed in the "Lexical Analyzer Development" laboratory work;
 - performing syntactic analysis (parsing) of the program represented by the token stream (the parsing algorithm is selected according to the variant);
 - constructing a parse tree;
 - generating tables for identifiers and various constants with full information necessary for code generation;
 - generating a listing of the input program including messages about lexical and syntactic errors.

The work is performed according to **Variant 12**.

Grammar according to the variant:

Variant 12

1. `<signal-program> --> <program>`
2. `<program> --> PROCEDURE <procedure-identifier>
 <parameters-list> ; <block> ;`
3. `<block> --> <declarations> BEGIN <statements-list>
 END`
4. `<declarations> --> <label-declarations>`
5. `<label-declarations> --> LABEL <unsigned-integer>
 <labels-list>; |
 <empty>`
6. `<labels-list> --> , <unsigned-integer> <labels-list>
 |
 <empty>`
7. `<parameters-list> --> (<variable-identifier>
 <identifiers-list>) |
 <empty>`
8. `<identifiers-list> --> , <variable-identifier>
 <identifiers-list> |
 <empty>`
9. `<statements-list> --> <statement> <statements-list>
 |
 <empty>`
10. `<statement> --> <unsigned-integer> : <statement> |
 GOTO <unsigned-integer> ; |
 RETURN ; |
 ; |
 ($ <assembly-insert-file-identifier> $)`
11. `<variable-identifier> --> <identifier>`
12. `<procedure-identifier> --> <identifier>`
13. `<assembly-insert-file-identifier> --> <identifier>`
14. `<identifier> --> <letter><string>`
15. `<string> --> <letter><string> |
 <digit><string> |
 <empty>`
16. `<unsigned-integer> --> <digit><digits-string>`
17. `<digits-string> --> <digit><digits-string> |
 <empty>`
18. `<digit> --> 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9`
19. `<letter> --> A | B | C | D | ... | Z`

Parsing algorithm: top-down parsing using the recursive descent algorithm.

Test cases:

1. True-тест1:

Input program:

```
PROCEDURE MIN;  
  BEGIN  
END;
```

Output program:

Result Lexem List:

Row	Col	Code	Lexem
-----	-----	------	-------

1	1	401	PROCEDURE
1	11	1001	MIN
1	14	59	;
2	5	402	BEGIN
3	1	403	END
3	4	59	;

Identifiers:

Code	Lexem	Type
------	-------	------

1001	MIN	-
------	-----	---

Constants:

Code	Lexem	Type
------	-------	------

Result Parse tree:

```
--<signal-program>  
|--<program>  
| |-- 401 PROCEDURE  
| |--<procedure-identifier>  
| | |-- 1001 <identifier>  
| | |--<parameters-list>  
| | |--<empty>  
| | -- 59 ;  
| |--<block>  
| | |--<declarations>  
| | | |--<label-declarations>  
| | | | |--<empty>  
| | |-- 402 BEGIN  
| | |--<statements-list>  
| | | |--<empty>  
| | |-- 403 END  
| | -- 59 ;
```

Identifiers:

Code	Lexem	Type
------	-------	------

1001	MIN	procedure-identifier
------	-----	----------------------

Constants:

Code	Lexem	Type
------	-------	------

Identifiers:

Code	Lexem	Type
------	-------	------

1001	MIN	procedure-identifier
------	-----	----------------------

Constants:

Code	Lexem	Type
------	-------	------

2. True-тест2:

Input program:

```
PROCEDURE MAIN(A, B);  
LABEL 10, 20;  
BEGIN  
  ($ ASMFILE $)  
  10: ;  
  GOTO 20;  
  RETURN;  
END;
```

Output program:

Result Lexem List:

Row	Col	Code	Lexem
1	1	401	PROCEDURE
1	11	1001	MAIN
1	15	40	(
1	16	1002	A
1	17	44	,
1	19	1003	B
1	20	41	)
1	21	59	;
2	1	404	LABEL
2	7	501	10
2	9	44	,
2	11	502	20
2	13	59	;
3	1	402	BEGIN
4	5	301	(\$
4	8	1004	ASMFILE
4	16	302	\$)
5	5	501	10
5	7	58	:
5	9	59	;
6	5	405	GOTO
6	10	502	20
6	12	59	;
7	5	406	RETURN
7	11	59	;
8	1	403	END
8	4	59	;

Identifiers:

Code	Lexem	Type
1001	MAIN	-
1002	A	-
1003	B	-
1004	ASMFILE	-

Constants:

Code	Lexem	Type
501	10	unsigned-integer
502	20	unsigned-integer

Result Parse tree:

```
--<signal-program>  
|--<program>  
| |-- 401 PROCEDURE  
| |--<procedure-identifier>  
| | |-- 1001 <identifier>  
| | |--<parameters-list>  
| | |-- 40 (  
| | |--<variable-identifier>  
| | | |-- 1002 <identifier>  
| | |--<identifiers-list>  
| | | |-- 44 ,  
| | |--<variable-identifier>
```

```

| | | | |-- 1003 <identifier>
| | | | |--<identifiers-list>
| | | | |--<empty>
| | |-- 41 )
| |-- 59 ;
| |--<block>
| |--<declarations>
| | |--<label-declarations>
| | | |-- 404 LABEL
| | | |-- 501 <unsigned-integer>
| | | |--<labels-list>
| | | | |-- 44 ,
| | | | |-- 502 <unsigned-integer>
| | | | |--<labels-list>
| | | | |--<empty>
| | | |-- 59 ;
| |-- 402 BEGIN
| |--<statements-list>
| | |--<statement>
| | | |-- 301 ($)
| | | |--<assembly-insert-file-identifier>
| | | | |-- 1004 <identifier>
| | | |-- 302 $)
| | |--<statements-list>
| | | |--<statement>
| | | | |-- 501 <unsigned-integer>
| | | | |-- 58 :
| | | | |--<statement>
| | | | |-- 59 ;
| | | |--<statements-list>
| | | | |--<statement>
| | | | |-- 405 GOTO
| | | | |-- 502 <unsigned-integer>
| | | | |-- 59 ;
| | | |--<statements-list>
| | | | |--<statement>
| | | | |-- 406 RETURN
| | | | |-- 59 ;
| | | |--<statements-list>
| | | | |--<empty>
| |-- 403 END
|-- 59 ;

```

Identifiers:

Code	Lexem	Type
1001	MAIN	procedure-identifier
1002	A	variable-identifier
1003	B	variable-identifier
1004	ASMFILE	assembly-insert-file-identifier

Constants:

Code	Lexem	Type
501	10	unsigned-integer
502	20	unsigned-integer

3. False-тест1:

Input program:

```
PROCEDURE TEST(X)
BEGIN
    RETURN;
END;
```

Output Program:

Result Lexem List:

Row	Col	Code	Lexem
1	1	401	PROCEDURE
1	11	1001	TEST
1	15	40	(
1	16	1002	X
1	17	41	)
2	1	402	BEGIN
3	5	406	RETURN
3	11	59	;
4	1	403	END
4	4	59	;

Identifiers:

Code	Lexem	Type
1001	TEST	-
1002	X	-

Constants:

Code	Lexem	Type
------	-------	------

Result Parse tree:

Parser: Error(2,1) <program> Not Single-character separator ';

Parser: Error(2,1) Tokens after the end of the program

```
--<signal-program>
|--<program>
| |-- 401 PROCEDURE
| |--<procedure-identifier>
| | |-- 1001 <identifier>
| | |--<parameters-list>
| | | |-- 40 (
| | | |--<variable-identifier>
| | | | |-- 1002 <identifier>
| | | |--<identifiers-list>
| | | | |--<empty>
| | | |-- 41 )
```

Identifiers:

Code	Lexem	Type
1001	TEST	procedure-identifier
1002	X	variable-identifier

Constants:

Code	Lexem	Type
------	-------	------

4. False-тест2:

Input program:

```
PROCEDURE FEST;  
BEGIN  
    LABEL 100;  
    RETURN;  
END;
```

Output program:

Result Lexem List:

Row	Col	Code	Lexem
1	1	401	PROCEDURE
1	11	1001	FEST
1	15	59	;
2	1	402	BEGIN
3	5	404	LABEL
3	11	501	100
3	14	59	;
4	5	406	RETURN
4	11	59	;
5	1	403	END
5	4	59	;

Identifiers:

Code	Lexem	Type
1001	FEST	-

Constants:

Code	Lexem	Type
501	100	unsigned-integer

Result Parse tree:

Parser: Error(3,5) <block> Not KEYWORD 'END'

Parser: Error(3,5) <program> Not Single-character separator ';'

Parser: Error(3,5) Tokens after the end of the program

```
--<signal-program>  
|--<program>  
| |-- 401 PROCEDURE  
| |--<procedure-identifier>  
| | |-- 1001 <identifier>  
| | |--<parameters-list>  
| | |--<empty>  
| |-- 59 ;  
| |--<block>  
| | |--<declarations>  
| | | |--<label-declarations>  
| | | | |--<empty>  
| | |-- 402 BEGIN  
| | |--<statements-list>  
| | | |--<empty>
```

Identifiers:

Code	Lexem	Type
1001	FEST	procedure-identifier

Constants:

Code	Lexem	Type
501	100	unsigned-intege

5. False-тест3:

Input program:

```
PROCEDURE FALSETEST(A; B);
  BEGIN
END;
```

Output program:

Result Lexem List:

Row	Col	Code	Lexem
1	1	401	PROCEDURE
1	11	1001	FALSETEST
1	20	40	(
1	21	1002	A
1	22	59	;
1	24	1003	B
1	25	41	)
1	26	59	;
2	5	402	BEGIN
3	1	403	END
3	4	59	;

Identifiers:

Code	Lexem	Type
1001	FALSETEST	-
1002	A	-
1003	B	-

Result Parse tree:

Parser: Error(1,22) <parameters-list> Not Single-character separator ')'

Parser: Error(1,24) <block> Not KEYWORD 'BEGIN'

Parser: Error(1,24) <program> Not Single-character separator ';'

Parser: Error(1,24) Tokens after the end of the program

--<signal-program>

```
--<program>
| --<program>
| | -- 401 PROCEDURE
| | --<procedure-identifier>
| | | -- 1001 <identifier>
| | --<parameters-list>
| | | -- 40 (
| | | --<variable-identifier>
| | | | -- 1002 <identifier>
| | | --<identifiers-list>
| | | | --<empty>
| | -- 59 ;
| | --<block>
| | | --<declarations>
| | | | --<label-declarations>
| | | | | --<empty>
```

Identifiers:

Code	Lexem	Type
1001	FALSETEST	procedure-identifier
1002	A	variable-identifier
1003	B	-